The documentation and process conversion measures necessary to comply with this revision shall be completed by 12 November 2018.

INCH-POUND

MIL-PRF-19500/525G
12 August 2018
SUPERSEDING
MIL-PRF-19500/525F
13 December 2013

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, POWER
TYPES 2N6546,AND 2N6547 JAN, JANTX, JANTXV AND JANHC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for NPN, silicon, power transistors. Three levels of product assurance (JAN, JANTX, and JANTXV) are provided for each device type and one level of product assurance (JANHC) is provided for each unencapsulated device type.

* 1.2 Package outlines. The device package outline is as follows: TO-3 in accordance with figure 1, TO-254 in accordance with figure 2, and TO-257 in accordance with figure 3 for all encapsulated device types. See figure 4 for unencapsulated devices.

1.3 Maximum ratings. Unless otherwise specified, $T_A$ = +25°C.

| Limits | $P_T$ (1) | | $V_{CEX}$ | $V_{CEO}$ | $V_{EBO}$ | $I_B$ | $I_C$ | $T_J$ and $T_{STG}$ | $R_{\theta JC}$ (2) |
|---|---|---|---|---|---|---|---|---|---|
| | $T_C$ = +25°C | $T_C$ = +100°C | | | | | | | |
| | W | W | V dc | V dc | V dc | A dc | A dc | °C | °C/W |
| 2N6546 | 175 | 100 | 600 | 300 | 8 | 10 | 15 | -65 to +200 | 1.0 |
| 2N6546T1 | 175 | 100 | 600 | 300 | 8 | 10 | 15 | -65 to +200 | 1.0 |
| 2N6546T3 | 125 | 100 | 600 | 300 | 8 | 10 | 15 | -65 to +200 | 1.4 |
| 2N6547 | 175 | 100 | 850 | 400 | 8 | 10 | 15 | -65 to +200 | 1.0 |
| 2N6547T1 | 175 | 100 | 850 | 400 | 8 | 10 | 15 | -65 to +200 | 1.0 |
| 2N6547T3 | 125 | 100 | 850 | 400 | 8 | 10 | 15 | -65 to +200 | 1.4 |

(1) See figure 5 for derating curves.
(2) See figures 6, 7, and 8 for thermal impedance graphs.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics.

| Limits | $h_{FE2}$ (1) | $h_{FE3}$ (1) | $C_{obo}$ | $\lvert h_{fe} \rvert$ | Switching times (2) | |
|---|---|---|---|---|---|---|
| | $V_{CE}$ = 2 V dc<br>$I_C$ = 5 A dc | $V_{CE}$ = 2 V dc<br>$I_C$ = 10 A dc | $V_{CB}$ = 10 V dc<br>$I_E$ = 0<br>0.1 MHz ≤ f ≤ 1 MHz | $V_{CE}$ = 10 V dc<br>$I_C$ = 0.5 A dc<br>f = 1 MHz | $V_{CC}$ = 250 V dc<br>$I_C$ = 10 A dc | |
| | | | | | $t_{on}$ | $t_{off}$ |
| | | | pF | | | |
| Min | 12 | 6 | | 6 | | |
| Max | 60 | | 500 | 30 | 1.0 | 4.7 |

(1) Pulsed (see 4.5.1).
(2) See figure 9 for pulse response circuits.

* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

* 1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, and "JANTXV".

* 1.5.2 JAN certification mark and quality level for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC".

* 1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

* 1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

* 1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "6546" and "6547".

* 1.5.3.3 Suffix letters. No suffix letters are used on devices package in the TO-3 package of figure 1. The suffix letters "T1" are used on devices that are packaged in the TO-254 package of figure 2. The suffix letters "T3" are used on devices that are packaged in the TO-257 package of figure 3.

* 1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

* 1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet is "A" (see figure 4 and 6.5).



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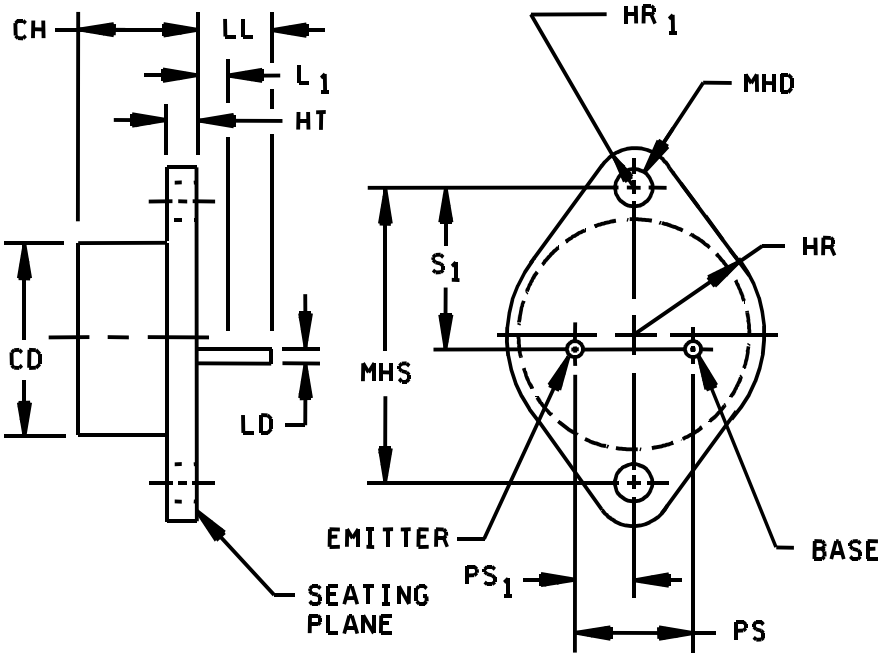



FIGURE 1. Dimensions and configuration (TO-3).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | | .875 | | 22.23 | 3 |
| CH | .250 | .450 | 6.35 | 11.43 | |
| HR | .495 | .525 | 12.57 | 13.34 | |
| $HR_1$ | .131 | .188 | 3.33 | 4.78 | 6 |
| HT | .050 | .135 | 1.27 | 3.43 | |
| LD | .038 | .043 | 0.97 | 1.09 | 4,5,9 |
| LL | .312 | | 7.92 | | 4,5,9 |
| $L_1$ | | .050 | | 1.27 | |
| MHD | .151 | .161 | 3.84 | 4.09 | 7 |
| MHS | 1.177 | 1.197 | 29.90 | 30.40 | |
| PS | .420 | .440 | 10.67 | 11.18 | 5 |
| $PS_1$ | .205 | .225 | 5.21 | 5.72 | 5 |
| S1 | .655 | .675 | 16.64 | 17.15 | |

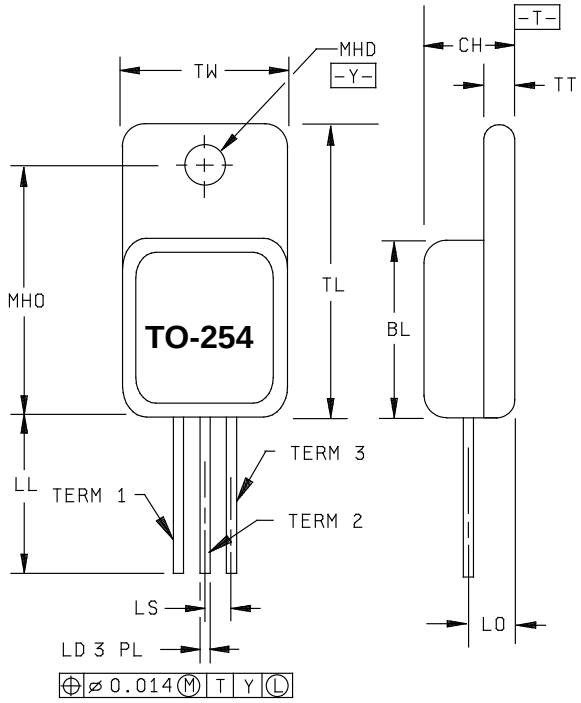
NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Body contour is optional within zone defined by CD.
4. These dimensions shall be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement shall be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. LD applies between L1 and LL. Lead diameter shall not exceed twice LD within L1.
10. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.
11. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to 0.006 inch (0.15 mm) convex overall.
12. Mounting holes shall be deburred on the seating plane side.

FIGURE 1. Dimensions and configuration (TO-3) - Continued.



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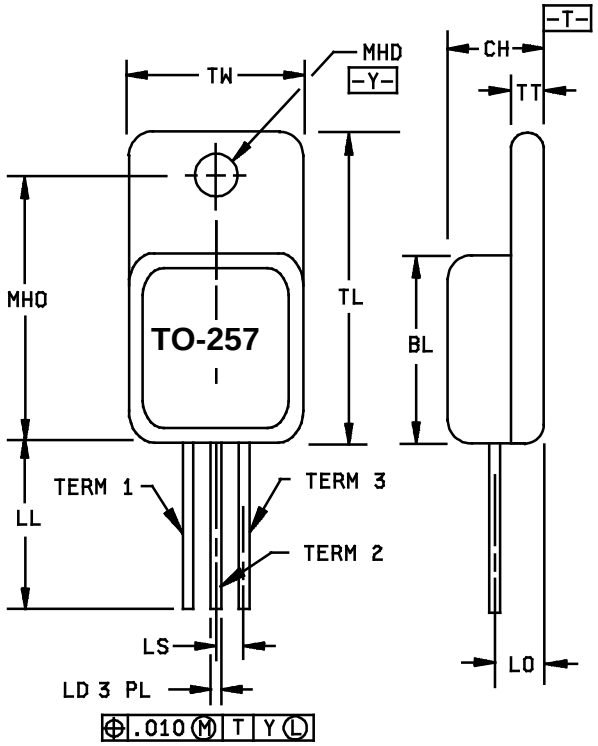

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .535 | .545 | 13.59 | 13.84 |
| CH | .249 | .260 | 6.32 | 6.60 |
| LD | .035 | .045 | 0.89 | 1.14 |
| LL | .510 | .570 | 13.46 | 13.97 |
| LO | .150 BSC | | 3.81 BSC | |
| LS | .150 BSC | | 3.81 BSC | |
| MHD | .139 | .149 | 3.53 | 3.78 |
| MHO | .665 | .685 | 16.89 | 17.40 |
| TL | .790 | .800 | 20.07 | 20.32 |
| TT | .040 | .050 | 1.02 | 1.27 |
| TW | .535 | .545 | 13.59 | 13.84 |
| Term 1 | Base | | | |
| Term 2 | Collector | | | |
| Term 3 | Emitter | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. All terminals are isolated from case.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Dimensions and configuration for 2N6546T1 and 2N6547T1 (TO-254).



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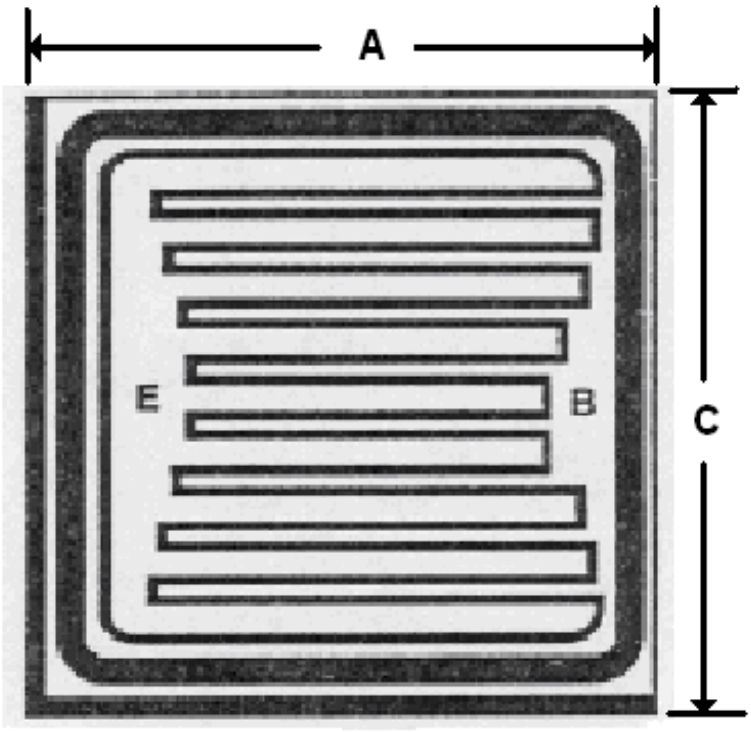

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .410 | .430 | 10.41 | 10.92 |
| CH | .190 | .200 | 4.83 | 5.08 |
| LD | .025 | .035 | 0.64 | 0.89 |
| LL | .500 | .750 | 12.70 | 19.05 |
| LO | .120 BSC | | 3.05 BSC | |
| LS | .100 BSC | | 2.54 BSC | |
| MHD | .140 | .150 | 3.56 | 3.81 |
| MHO | .527 | .537 | 13.39 | 13.63 |
| TL | .645 | .665 | 16.38 | 16.89 |
| TT | .035 | .045 | 0.89 | 1.14 |
| TW | .410 | .420 | 10.41 | 10.67 |
| Term 1 | Base | | | |
| Term 2 | Collector | | | |
| Term 3 | Emitter | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Methods used for electrical isolation of the terminals feedthroughs shall employ materials that contain a minimum of 90 percent AL2O3 (ceramic).
4. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 3. Dimensions and configuration for 2N6546T3 and 2N6547T3 (TO-257).



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| Ltr. | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A, C | .205 | .215 | 5.21 | 5.46 |

DESIGN DATA

Metalization:

| | |
|---|---|
| Top: | Aluminum 54,000 Å minimum, 60,000 Å nominal. |
| Back: | Al/Ti/Ni/Au 10,000 Å minimum, 12,500 Å nominal. |
| Back side: | Collector. |
| Chip thickness: | .012 inch (0.305 mm) ±.002 inch (0.051 mm). |
| Bonding pad: | B = .018 inch (0.46 mm) x .040 (1.02 mm).<br>E = .018 inch (0.46 mm) x .040 (1.02 mm). |

* FIGURE 4. JANHCA (A version) die dimensions.



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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-3), figure 2 (TO-254), figure 3 (TO-257) and figure 4 (JANHCA).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition requirements (see 6.2).

3.4.2 Construction. These devices shall be constructed in a manner and using materials which enable the devices to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.



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3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC qualification. JANHC qualification inspection shall be in accordance with MIL-PRF-19500.



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* 4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| | Screen (see table E-IV of MIL-PRF-19500) | Measurements<br>JANTX, JANTXV levels |
|---|---|---|
| | (1) 3c | Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.2). |
| | 10 | $T_A$ = +150°C; t = 48 hours min.<br>$V_{CB}$ = 200 V dc for 2N6546, 2N6546T1, 2N6546T3.<br>$V_{CB}$ = 300 V dc for 2N6547, 2N6547T1, 2N6547T3. |
| | 11 | $h_{FE2}$ ; $I_{CEX1}$ |
| | 12 | See 4.3.1. |
| | 13 | Subgroup 2 of table I herein.<br>$\Delta I_{CEX1}$ = 100 percent of initial value or 2 μA dc, whichever is greater;<br>$\Delta h_{FE2}$ = ±25 percent of initial value. |
| * | 17 | For TO-254 and TO-257 packages: Method 1081 of MIL-STD-750 (see 4.3.4), Endpoints: Subgroup 2 of table I herein. |

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $T_J$ = + 175°C minimum, $V_{CB} \geq$ 20 V dc, $T_A$ = +35°C maximum.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$, $t_{SW}$, (and $V_H$ where appropriate). See table II, group E, subgroup 4 herein.

4.3.3 Screening (JANHC). Screening of JANHC die shall be in accordance with MIL-PRF-19500, “Discrete Semiconductor Die/Chip Lot Acceptance”. Burn-in duration for the JANHC level follows JANTX requirements.

4.3.4 Dielectric withstanding voltage.

a. Magnitude of test voltage……..900 V dc.

b. Duration of application of test voltage……..15 seconds (min).

c. Points of application of test voltage……..All leads to case (bunch connection).

d. Method of connection……..Mechanical.

e. Kilovolt-ampere rating of high voltage source…….1,200 V/1.0 mA (min).

f. Maximum leakage current……..1.0 mA.

g. Voltage ramp up time.....500 V/second.



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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB of MIL-PRF-19500 and herein.

* 4.4.2.1 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 1037 | For solder die attach: $V_{CB} \geq 20$ V dc, for 2,000 cycles, $T_A \leq +35°C$, adjust power or current to achieve a minimum $\Delta T_J = +100°C$. |
| B3 | 1027 | For eutectic die attach: $V_{CB} \geq 20$ V dc; adjust $P_T$ to achieve $T_J = +175°C$ minimum, $T_A \leq +35°C$. |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Test condition A; weight = 10 pounds; t = 15 seconds. |
| C5 | 3131 | See 4.3.2, $R_{\theta JC} = 1.0°C/W$ for 2N6546, 2N6546T1, 2N6547 and 2N6547T1; $R_{\theta JC} = 1.4°C/W$ for 2N6546T3 and 2N6547T3. |
| C6 | 1037 | For solder die attach: $V_{CB} \geq 20$ V dc, for 6,000 cycles, $T_A \leq +35°C$, adjust power or current to achieve a minimum $\Delta T_J = +100°C$. |
| C6 | 1026 | For eutectic die attach: $V_{CB} \geq 20$ V dc; adjust $P_T$ to achieve $T_J = +175°C$ minimum, $T_A \leq +35°C$. |

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 2/ | 3131 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |
| Collector to emitter breakdown voltage | 3011 | Bias condition D; $I_C$ = 100 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | | | |
| 2N6546, 2N6546T1, 2N6546T3<br>2N6547, 2N6547T1, 2N6547T3 | | | | 300<br>400 | | V dc<br>V dc |
| Emitter to base cutoff current | 3061 | Bias condition D;<br>$V_{EB}$ = 8 V dc | $I_{EBO}$ | | 1.0 | mA dc |
| Collector to emitter cutoff current | 3041 | Bias condition A;<br>$V_{BE}$ = 1.5 V dc; | $I_{CEX1}$ | | 200 | µA dc |
| 2N6546, 2N6546T1, 2N6546T3<br>2N6547, 2N6547T1, 2N6547T3 | | $V_{CE}$ = 600 V dc<br>$V_{CE}$ = 850 V dc | | | | |
| Base emitter voltage (saturated) | 3066 | Test condition A;<br>$I_C$ = 10 A dc;<br>$I_B$ = 2.0 A dc; pulsed (see 4.5.1) | $V_{BE(sat)}$ | | 1.6 | V dc |
| Collector to emitter saturated voltage | 3071 | $I_C$ = 10 A dc; $I_B$ = 2.0 A dc; pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 1.5 | V dc |
| Collector to emitter saturated voltage | 3071 | $I_C$ = 15 A dc; $I_B$ = 3.0 A dc; pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 5.0 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc; $I_C$ = 1 A dc; pulsed (see 4.5.1) | $h_{FE1}$ | 15 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc; $I_C$ = 5 A dc; pulsed (see 4.5.1) | $h_{FE2}$ | 12 | 60 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc; $I_C$ = 10 A dc; pulsed (see 4.5.1) | $h_{FE3}$ | 6 | | |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A; $V_{BE}$ = 1.5 V dc | $I_{CEX2}$ | | 30 | mA dc |
| 2N6546, 2N6546T1, 2N6546T3 | | $V_{CE}$ = 600 V dc | | | | |
| 2N6547, 2N6547T1, 2N6547T3 | | $V_{CE}$ = 850 V dc | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A; $V_{BE}$ = 1.5 V dc; | $I_{CEX3}$ | | 450 | µA dc |
| 2N6546, 2N6546T1, 2N6546T3 | | $V_{CE}$ = 300 V dc | | | | |
| 2N6547, 2N6547T1, 2N6547T3 | | $V_{CE}$ = 400 V dc | | | | |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer current | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 5 A dc; pulsed (see 4.5.1) | $h_{FE4}$ | 4 | | |
| Subgroup 4 | | | | | | |
| Pulse response | 3251 | Test condition A except test circuit and pulse requirements in accordance with figure 9 herein. | | | | |
| Turn-on time | | $V_{CC}$ = 250 V dc; $I_C$ = 10 A dc; $I_{B1}$ = $I_{B2}$ = 2 A dc | $t_{on}$ | | 1.0 | µs |
| Turn-off time | | $V_{CC}$ = 250 V dc; $I_C$ = 10 A dc; $I_{B1}$ = $I_{B2}$ = 2 A dc | $t_{off}$ | | 4.7 | µs |
| Magnitude of common emitter small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 10 V dc; $I_C$ = .5 A dc; f = 1 MHz | $\lvert h_{fe} \rvert$ | 6.0 | 30 | |
| Output capacitance (open circuit) | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 0.1 MHz ≤ f ≤ 1.0 MHz | $C_{obo}$ | | 500 | pF |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area (dc operation) | 3051 | $T_C$ = +25°C; t = 1 s; 1 cycle; (see figure 10) | | | | |
| Test 1 (All device types) | | $I_C$ = 15 A dc, $V_{CE}$ = 11.7 V dc | | | | |
| Test 2 (All device types) | | $I_C$ = 8.75 A dc, $V_{CE}$ = 20 V dc | | | | |
| Test 3 2N6546, 2N6546T1, 2N6546T3 | | $V_{CE}$ = 250 V dc; $I_C$ = 45 mA dc | | | | |
| 2N6547, 2N6547T1, 2N6547T3 | | $V_{CE}$ = 350 V dc; $I_C$ = 30 mA dc | | | | |
| Safe operating area | 3053 | Load condition C; (unclamped inductive load); see figure 11; $T_C$ = +25°C; duty cycle ≤ 10 percent $R_S$ = 0.1 Ω; $t_r = t_f \leq 500$ ns | | | | |
| Test 1 | | $t_p$ = 5 ms; (vary to obtain $I_C$); $R_{BB1}$ = 15 Ω; $V_{BB1}$ = 38.5 V dc; $R_{BB2}$ = 50 Ω; $V_{BB2}$ = -4 V dc; $V_{CC}$ = 20 V dc; $I_C$ = 15 A dc; L = 10 µH 3/ | | | | |
| Test 2 | | $t_p$ = 5 ms; (vary to obtain $I_C$); $R_{BB1}$ = 15 Ω; $V_{BB1}$ = 38.5 V dc; $R_{BB2}$ = 50 Ω; $V_{BB2}$ = -4 V dc; $V_{CC}$ = 20 V dc; $I_C$ = 100 mA dc; L = 1 mH 4/ | | | | |
| Electrical measurements: | | | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A, $V_{BE}$ = 1.5 V dc | $I_{CEX3}$ | | | |
| 2N6546, 2N6546T1, 2N6546T3 | | $V_{CE}$ = 600 V dc | | | 2.0 | mA dc |
| 2N6547, 2N6547T1, 2N6547T3 | | $V_{CE}$ = 850 V dc | | | 2.0 | mA dc |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5- continued. | | | | | | |
| Safe operating area (switching) | | Clamped inductive load; $T_A$ = +25°C; duty cycle ≤ 5 percent; $t_p$ = 1.5 ms; (vary to obtain $I_C$); $V_{CC}$ = 20 V dc; $I_C$ = 8 A dc; L = 180 µH (see figure 12) | | | | |
| 2N6546, 2N6546T1, 2N6546T3 | | Clamp voltage = 350 V dc | | | | |
| 2N6547, 2N6547T1, 2N6547T3 | | Clamp voltage = 450 V dc | | | | |
| Electrical measurements | | Same as safe operating area (unclamped inductive) above | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.

* 2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

3/ L = 10 µH (approx. 10 turns, 1 row of #16ASG wire on an air core 2.875 inches (73.03 mm) ID) .0007 ohms, or equivalent.

4/ L = 1 mH (one each Miller type 7827 in parallel with two each series strung Miller type 7825 and this in series with two each series strung Miller type 7827) .45 ohms, or equivalent.



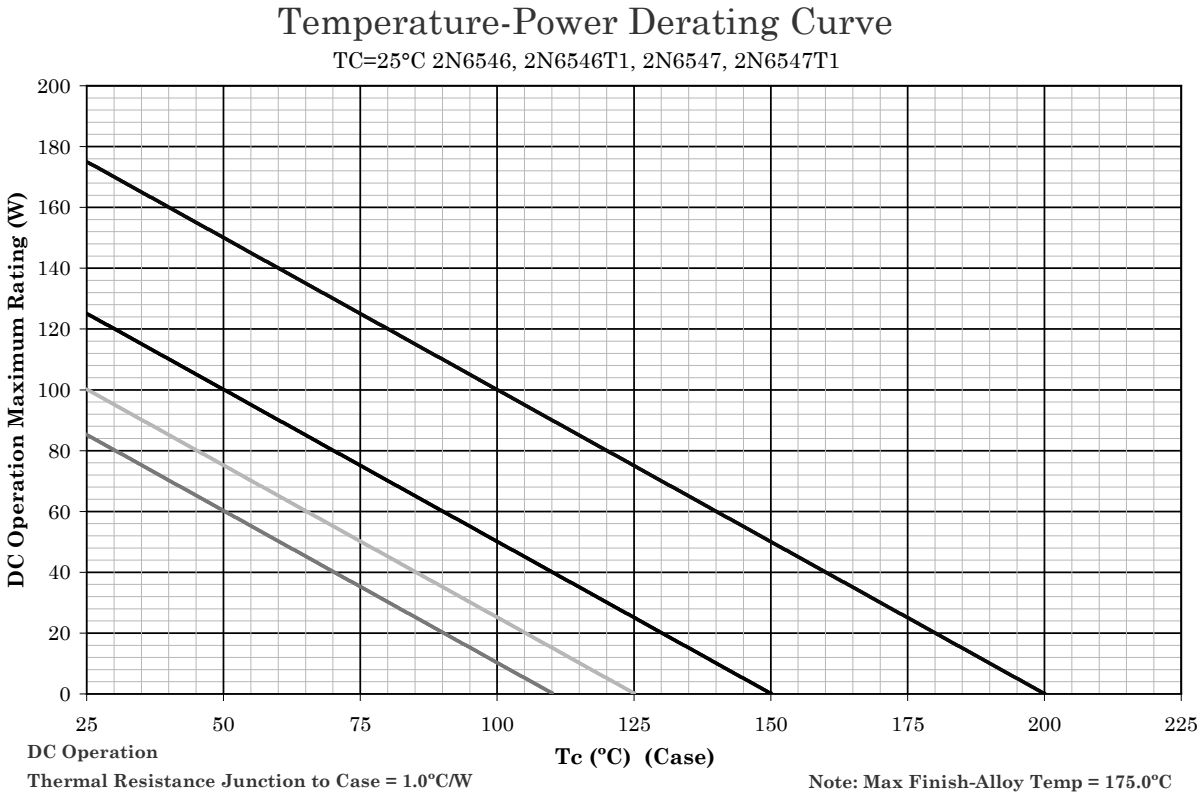
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* TABLE II. Group E inspection (all quality levels) - for qualification and re-qualification only.

| Inspection | MIL-STD-750 | | Sample size |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling | 1051 | | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Blocking life | 1048 | Test temperature = +125°C; $V_{CB}$ = 80 percent of rated; T = 1,000 hours. | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 4 | | | N/A |
| Thermal impedance curves | | See MIL-PRF-19500, table E-IX, group E, subgroup 4. | |
| * Subgroup 5 | | | 3 devices,<br>c = 0 |
| Barometric pressure | 1001 | Pressure = 8.0 mm Hg time = 60 seconds; normal mounting, see 1.4. | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition A for devices ≥ 400 V, condition B for devices < 400 V. | |



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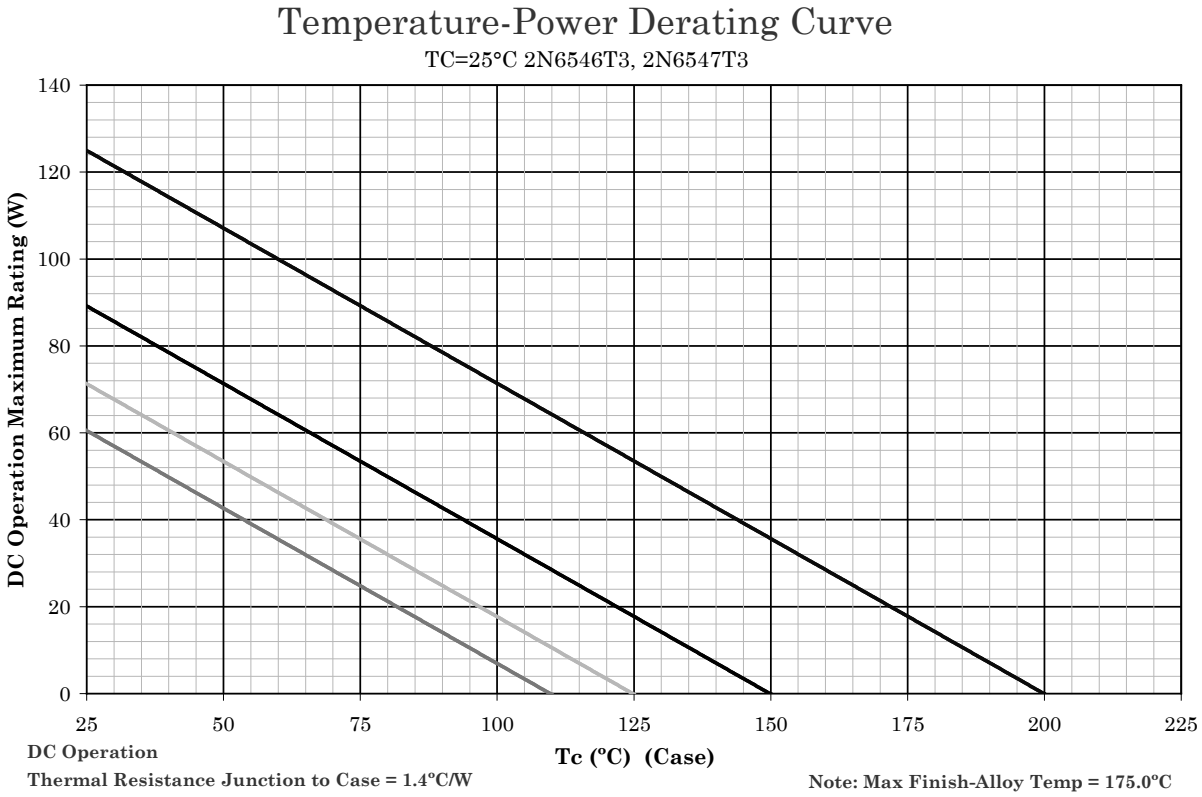

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ +200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ +125°C and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 5. Temperature-power derating graph.



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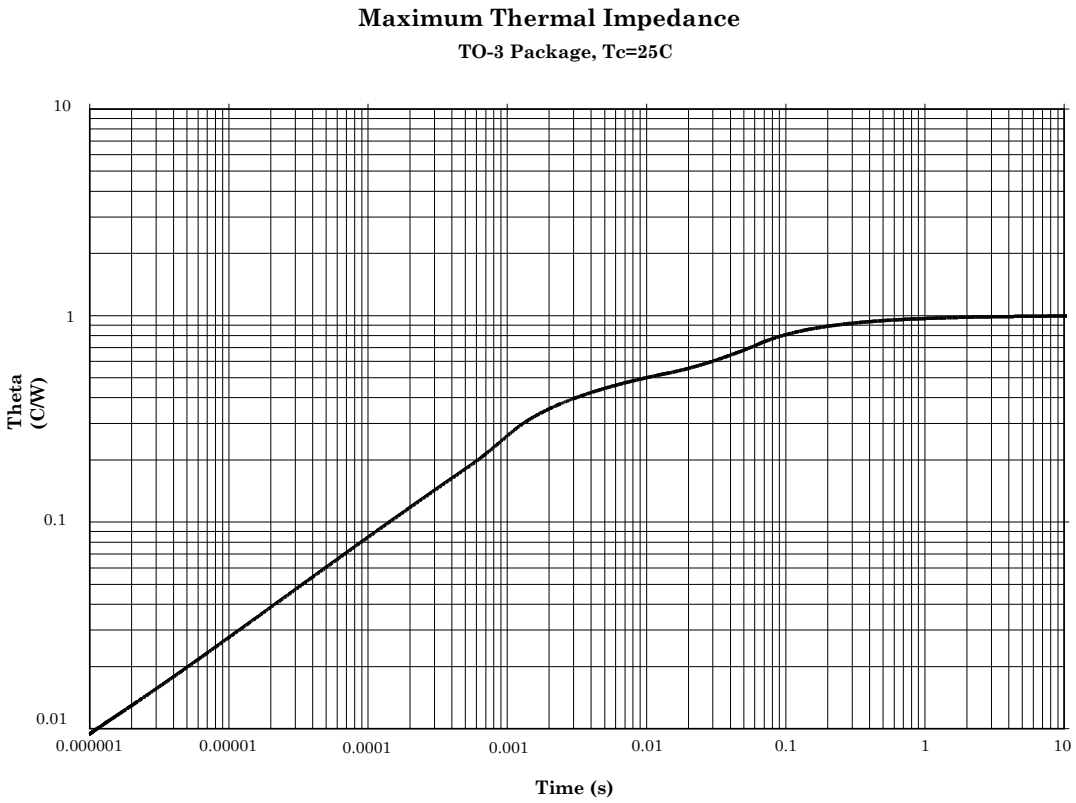

NOTES:

1. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ +200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ +150°C where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤ +125°C and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 5. Temperature-power derating graph - Continued.



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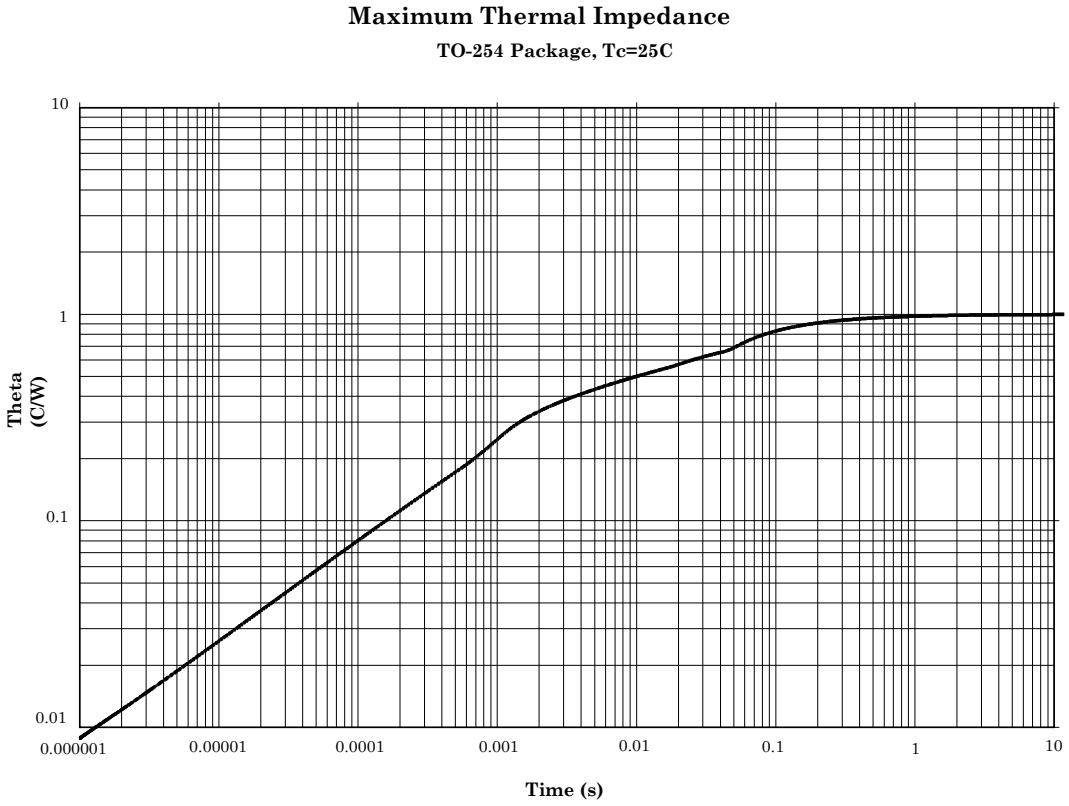



$R_{\theta JC}$ = 1.0 °C/W max., TO-3.

FIGURE 6. Thermal impedance graph (2N6546 and 2N6547).



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$R_{\theta JC}$ = 1.0 °C/W max.

FIGURE 7. Thermal impedance graph (2N6546T1 and 2N6547T1).



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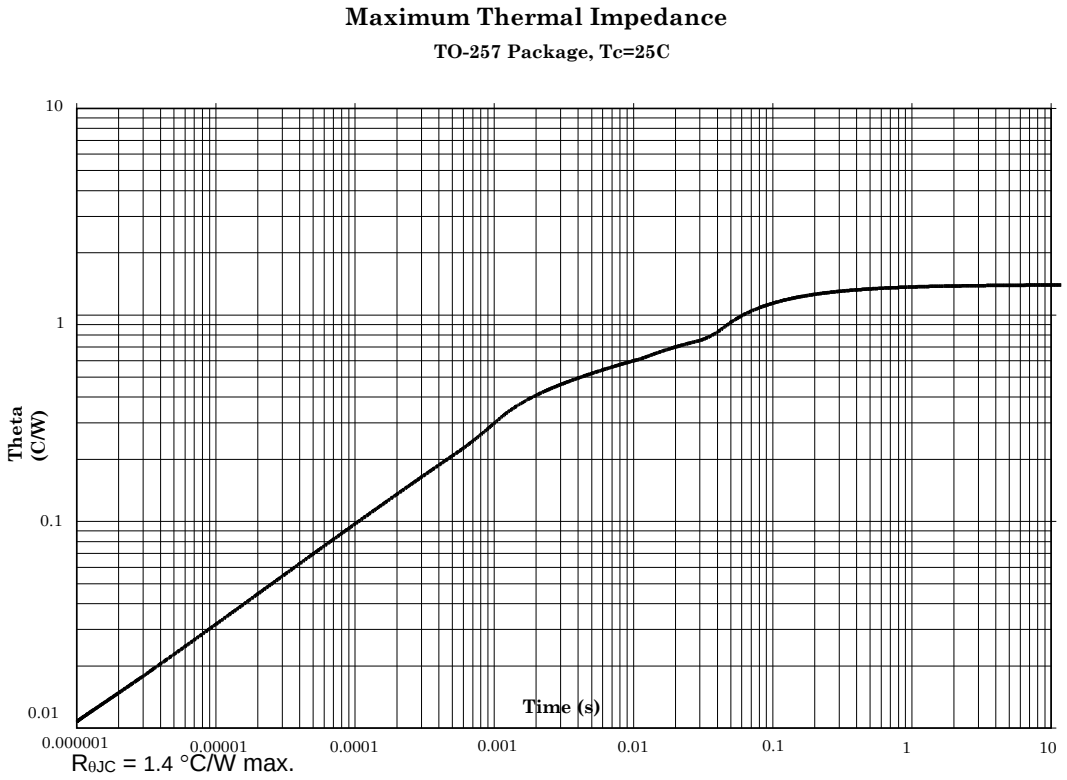
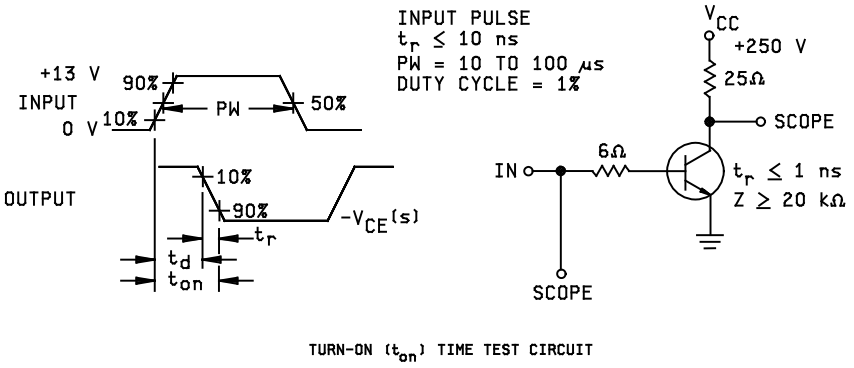


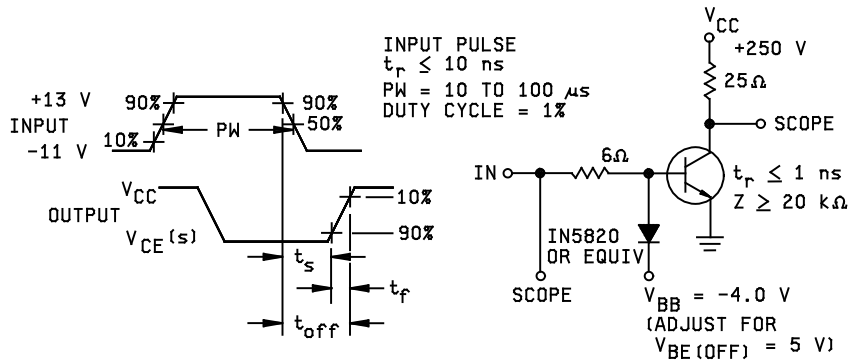

FIGURE 8. Thermal impedance graph (2N6546T3 and 2N6547T3).



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TURN-ON ($t_{on}$) TIME TEST CIRCUIT




TURN-OFF ($t_{off}$) TIME TEST CIRCUIT

FIGURE 9. Switching time test circuits.



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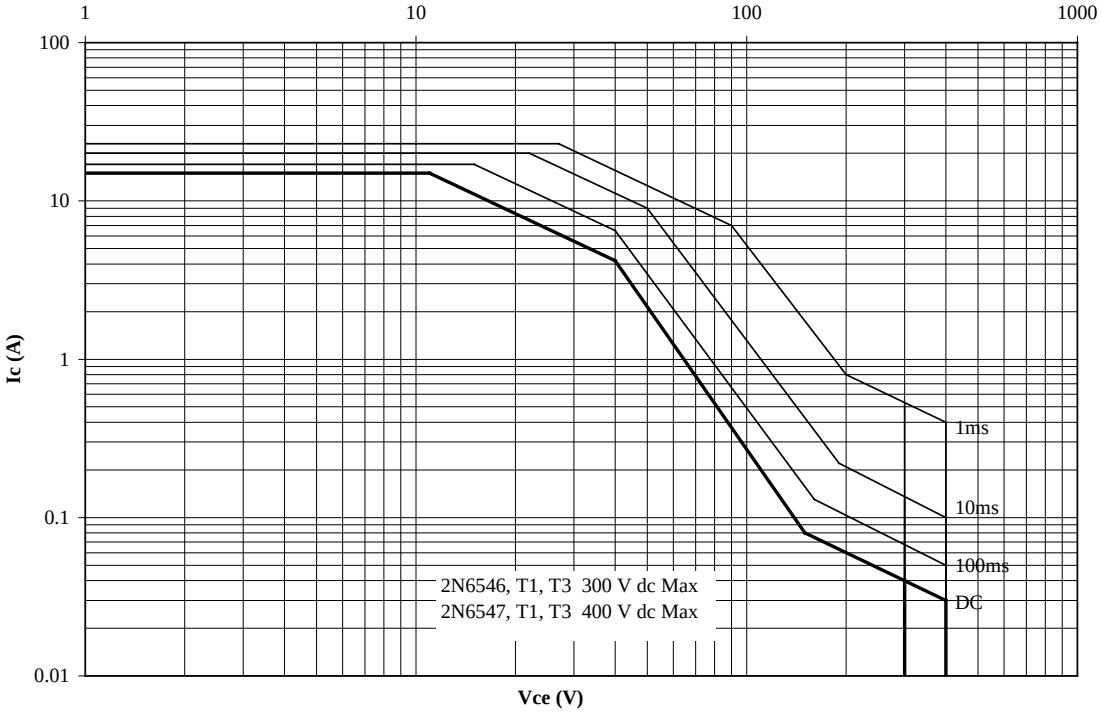



FIGURE 10. Maximum safe operating area graph (continuous dc).



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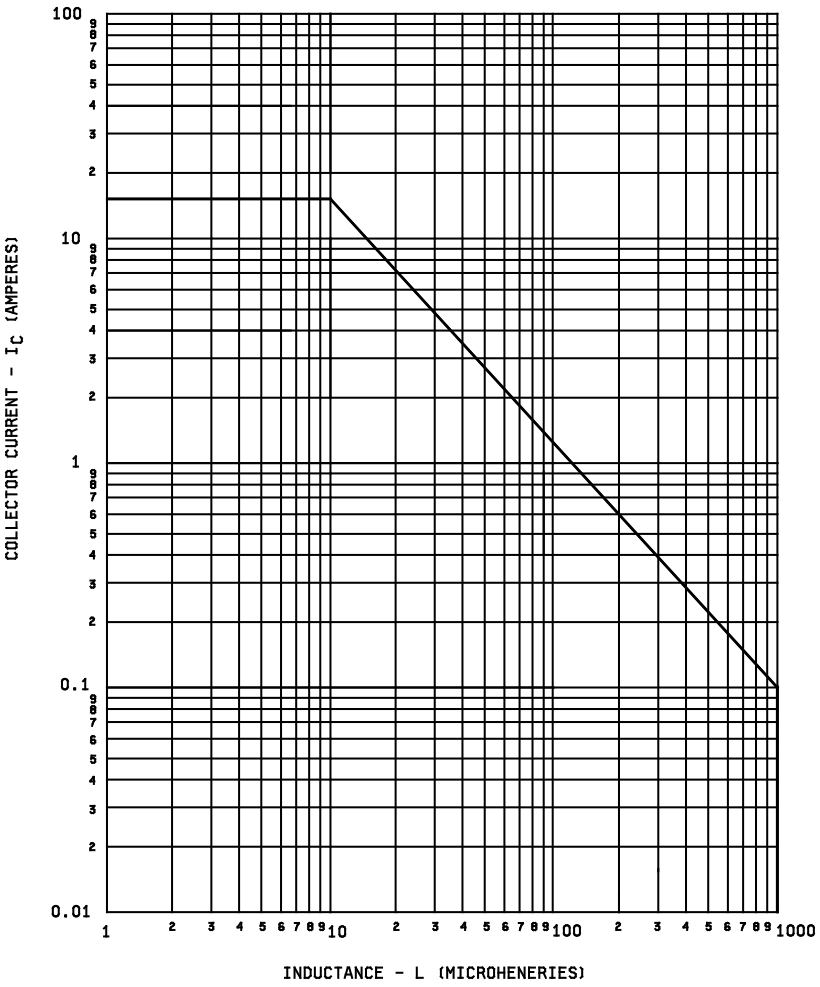
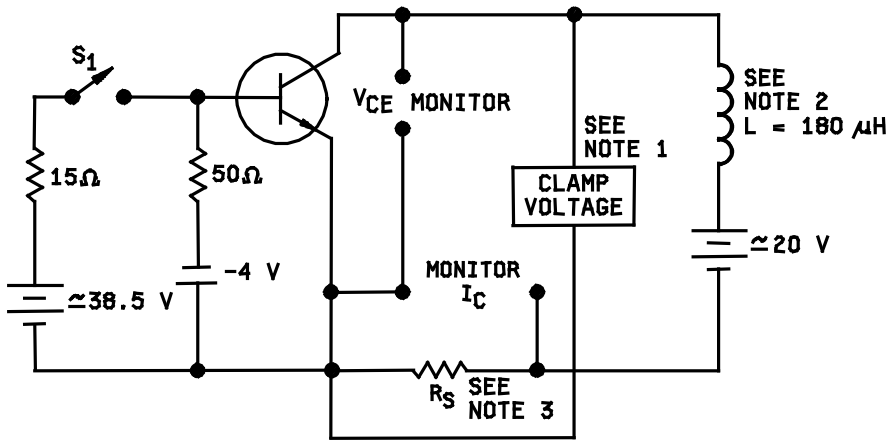



FIGURE 11. Safe operating area for switching between saturation and cutoff (unclamped inductive load, all devices).



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NOTES:

1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 180 μH at 8 A with a maximum dc resistance of 0.05 ohm. For reference only: Two each Miller type 7827 in parallel, or equivalent.
3. $R_S \leq .1$ ohm, 12 W, 1 percent tolerance maximum, (non-inductive).

Procedure:

1. With switch $S_1$ closed, set the specified test conditions.
2. Open $S_1$. Device fails if clamp voltage not reached and maintained until current reaches zero.
3. Perform specified end-point tests.

FIGURE 12. Clamped inductive sweep test circuit.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

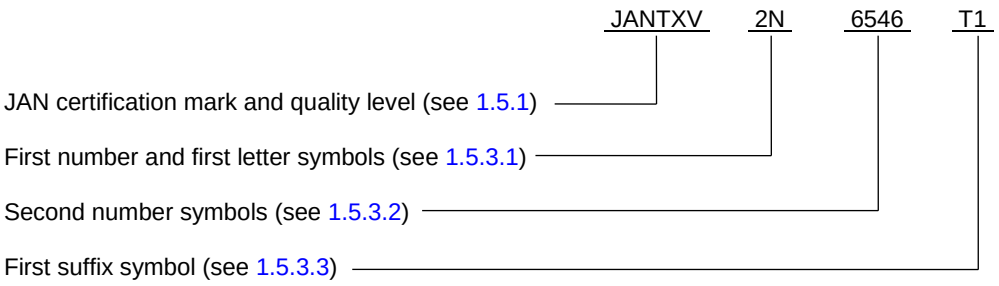
c. Lead finish (see 3.4.1).

* d. The complete PIN, see 1.5 and 6.5.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://qpldocs.dla.mil.

* 6.4 PIN construction example.

* 6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.






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* 6.5 List of PINs.

* 6.5.1 List of PINs. The following is a list of possible PINs available on this specification sheet.

| PINs for types2N6546 | PINs for types2N6547 |
|---|---|
| JAN2N6546 | JAN2N6547 |
| JAN2N6546T1 | JAN2N6547T1 |
| JAN2N6546T3 | JAN2N6547T3 |
| JANTX2N6546T1 | JANTX2N6547T1 |
| JANTX2N6546T3 | JANTX2N6547T3 |
| JANTX2N6546T1 | JANTX2N6547T1 |
| JANTXV2N6546T1 | JANTXV2N6547T1 |
| JANTXV2N6546T3 | JANTXV2N6547T3 |
| JANTXV2N6546T3 | JANTXV2N6547T3 |

* 6.5.2 List of PINs for unencapsulated devices. The following is a list of possible PINs available on this specification sheet. The qualified JANHC suppliers with the applicable letter version (example JANHCA2N6546) will be identified on the QML.

| JANHC ordering information | |
|---|---|
| PIN | Manufacturer<br>43611 |
| 2N6546<br>2N6547 | JANHCA2N6546<br>JANHCA2N6547 |

* 6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2018-071)

Review activities:
Army - AR, MI
Air Force - 19, 99

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